COMP9601 26s1 Final Practice

W1 Intro and System View

Q1. MCQ
Which statement best captures the stored-program idea?

(A) Programs are kept only in CPU registers.
(B) Instructions and data are both represented as bit patterns in addressable memory.
(C) The operating system executes without using memory.
(D) Network protocols decide the instruction format.

Q2. Short Answer Pitfall
Distinguish latency from throughput using a CPU-and-disk example.

Q3. Calculation
A trace reports 2.5 million instructions executed in 0.5 ms. Compute the instruction throughput in MIPS.

Q4. True/False Warning
True or false: a byte-addressed machine with a 32-bit address field can address exactly 2^{32} words. Explain.

Q5. Short Answer Pitfall
Why is unit checking especially important in COMP9601 performance questions?

Q6. MCQ
A program spends most of its time waiting for storage. Which observation is most defensible?

(A) Increasing CPU clock always halves total execution time.
(B) The bottleneck is likely I/O or memory rather than arithmetic.
(C) The program must use the wrong number base.
(D) Packet propagation delay is the main cause.

Q7. Short Answer
State two assumptions behind treating a computer as CPU, memory and I/O blocks.

Q8. Calculation
A 64-bit word contains how many bytes and nibbles?

Q9. Short Answer
Give one reason a correct calculation may still lose marks in an exam answer.

Q10. Long Question Strategy
A student writes: *A computer is fast if its CPU clock is high.*

(a) Identify two missing system components in this claim.
(b) Give a counterexample where higher CPU speed gives little total improvement.
(c) State one metric for latency and one for throughput.
(d) Explain how this mindset connects to Amdahl's law later in the course.

W2 Radix and Integer Representation

Q1. Calculation
Convert $(187)_{10}$ to base 3 using the division-remainder method.

Q2. Calculation **Semester 1 2022 Final (modified)**
Convert $(34)_8$ to base 4.

Q3. MCQ **Pitfall**
Which 8-bit two's-complement pattern represents -25?

(A) 00011001
(B) 10011001
(C) 11100111
(D) 11101001

Q4. Short Answer
Why do signed magnitude and one's complement have two representations of zero?

Q5. Calculation **Formula**
What are the minimum and maximum signed decimal values in a 12-bit two's-complement word?

Q6. True/False **Warning**
True or false: if an 8-bit unsigned addition has a carry out, then the same bit pattern must be an overflow in 8-bit two's complement.

Q7. Short Answer
Explain the difference between truncation and overflow.

Q8. Calculation
A 16-bit memory word stores two bytes. How many addressable bytes are in 2^{20} words?

Q9. Short Answer **Pitfall**
In little-endian order, how is the 32-bit value 0x0003A74B stored starting at address 0x134A?

Q10. Long Question **Semester 1 2024 Final (modified)**
A 6-bit two's-complement adder adds $+17$ and -25.

(a) Encode both operands.
(b) Perform the bit addition.
(c) Give the signed decimal result.
(d) State whether signed overflow occurred and why.

W3 Floating Point and Error Codes

Q1. Short Answer Pitfall
Why does finite significand length cause numerical error?

Q2. Calculation
For a floating format with 5 exponent bits, what is the usual exponent bias?

Q3. MCQ Semester 1 2023 Final (modified)
Which method is best suited to detecting burst errors in transmitted frames?

(A) A single parity bit only
(B) CRC with a suitable generator polynomial
(C) Changing to signed magnitude
(D) Increasing CPU registers

Q4. Calculation Formula
Using even parity, what parity bit is appended to data 1011010?

Q5. Short Answer Warning
In CRC division, why are XOR operations used instead of ordinary subtraction?

Q6. Calculation Formula
Generator 1001 is used for CRC. How many zeros are appended before division?

Q7. True/False
True or false: ASCII 0x48 0x65 0x6C 0x6C 0x6F decodes to Hello.

Q8. Short Answer
Compare parity and Hamming-style checking for error handling.

Q9. Calculation
A 10-bit fraction field has an implicit leading 1 in normalized binary. How many explicit fraction bits are stored?

Q10. Long Question Semester 1 2021 Final (modified)
A sender transmits data bits 1101011 using CRC generator 1001.

(a) State the degree of the generator.
(b) Describe exactly what is appended before division.
(c) Explain how the receiver tests the received codeword.
(d) If the receiver's division remainder is nonzero, what conclusion is justified?

W4 Boolean Algebra and Logic Gates

Q1. Calculation
Construct the truth-table output column for $F(x,y,z) = xy + xz$ in row order 000 to 111.

Q2. Calculation Formula
Simplify $x + xy$ and name the identity.

Q3. MCQ Pitfall
Which identity is most directly used in $(xy)' = x' + y'$?

(A) Idempotent law
(B) Absorption law
(C) De Morgan's theorem
(D) Associative law

Q4. Short Answer
Why are NAND and NOR called universal gates?

Q5. Calculation
Simplify $x(y + y') + xz$.

Q6. True/False Warning
True or false: XOR and OR have the same truth table except when both inputs are 1.

Q7. Calculation Semester 1 2021 Final (modified)
For $F = (x + y' + z)(x' + y)(y + z')$, how many truth-table rows are required?

Q8. Short Answer Strategy
Give a reliable procedure for translating a logic-gate diagram into a Boolean expression.

Q9. Short Answer
What is the difference between SOP and POS forms?

Q10. Long Question
For $F(x,y,z) = xyz + xy + xz$:

(a) Factor the expression.
(b) Simplify it as far as possible.
(c) Identify one input row where $F = 1$.
(d) Explain why a truth table can verify the simplification.

W5 Simple Computer and Stack Expressions

Q1. MCQ Semester 1 2021 Final (modified)
For one-address instructions with 75 distinct operations, how many opcode bits are needed?

(A) 6
(B) 7
(C) 8
(D) 75

Q2. Calculation Semester 1 2022 Final (modified)
A 16-bit instruction has 61 operations and one address field. How many bits remain for the address?

Q3. Short Answer Pitfall
Name the roles of PC, IR, MAR and MBR/MDR.

Q4. Calculation
Convert infix $(A + B) * (C - D/E)$ to postfix.

Q5. Calculation Warning
Convert postfix $AB + CDE/ - *$ to fully parenthesised infix.

Q6. True/False
True or false: in a stack architecture, most arithmetic operands are implicit on the top of the stack.

Q7. Short Answer Formula
Why does exponentiation right-associativity matter in infix-to-postfix conversion?

Q8. Calculation
A 12-bit address field can address how many memory words?

Q9. Short Answer
Explain the difference between an instruction's opcode and operand/address field.

Q10. Long Question Pitfall
A machine has 16-bit instructions, 4 opcode bits and a 12-bit address/constant field.

(a) How many distinct opcodes are possible?
(b) Encode LOAD 0x400 if LOAD opcode is 0001.
(c) Encode HALT if opcode is 1111 and address field is 0.
(d) Explain why the hex representation is convenient here.

W6 Instruction Processing and Machine Code

Q1. **Short Answer** **Strategy**

List the main stages in the fetch-decode-execute cycle.

Q2. **MCQ**

Which event is most naturally handled by an interrupt?

(A) A Boolean expression being idempotent
(B) An I/O device becoming ready
(C) A hex digit becoming invalid
(D) A cache tag being longer

Q3. **Calculation**

Given opcode ADD=0011 and address 0x402, write the 16-bit hex instruction.

Q4. **Calculation** **Pitfall**

Given binary 0101 0001 0000 0000 in the simple machine, identify opcode and address.

Q5. **Short Answer** **Warning**

Why must a program that tests $X - LIMIT = 0$ be careful after executing SUB LIMIT?

Q6. **Calculation**

For the loop $sum = 1 + \cdots + 10$, what final decimal value should be stored?

Q7. **True/False**

True or false: a trap is a software-generated interrupt.

Q8. **Short Answer**

Why is using symbolic labels safer than writing jump addresses immediately?

Q9. **Calculation**

If a program starts at 0x100 and has 10 instructions before data begins, what is the address of the last instruction?

Q10. **Long Question** **Formula**

Design the control outline for $SUM = 0$; $X = 1$; while $X \leq 10$, $SUM = SUM + X$, $X = X + 1$.

(a) State suitable data labels and initial values.
(b) Give the high-level sequence of accumulator instructions.
(c) Explain why LIMIT can be 11.
(d) State the final value of SUM.

W7 ISA, Addressing Modes and Pipelining

Q1. MCQ Semester 1 2023 Final (modified)
Which architecture style is usually most pipeline-friendly for simple embedded tasks?

(A) Highly irregular CISC with variable-length instructions
(B) RISC with fixed-length simple instructions
(C) Decimal-only stack machine
(D) Manual packet switching

Q2. Calculation Pitfall
Instruction Load 500 is used. Memory[500]=200, Memory[200]=900 and R1=300. For direct addressing, what is the effective address and what value is loaded?

Q3. Calculation
Using the same data, what value is loaded by indirect addressing?

Q4. Calculation
Using the same Load 500 and R1=300, what address is used by indexed addressing?

Q5. Short Answer Strategy
Explain immediate addressing and give one advantage.

Q6. True/False Warning
True or false: pipelining always reduces the latency of a single instruction.

Q7. Calculation
Five instructions pass through a 4-stage balanced pipeline with 2 ns per stage. Estimate total time.

Q8. Short Answer Strategy
Name three pipeline hazards and one mitigation for any one of them.

Q9. Short Answer
Why might a CISC instruction be harder to pipeline than a RISC instruction?

Q10. Long Question
A nonpipelined processor has four stages F, D, E and W, each 3 ns.

(a) How long for 20 instructions without pipelining?
(b) Estimate ideal pipelined time.
(c) Compute speedup.
(d) Explain one reason real speedup may be lower.

W8 Memory Hierarchy and Cache

Q1. Calculation **Semester 1 2021 Final (modified)**
A direct-mapped cache has 256 lines, 64-byte lines and 2^{30} bytes of memory. Find offset, line and tag bits.

Q2. Calculation **Semester 1 2024 Final (modified)**
A 4 MiB cache has 512-byte blocks and is 16-way set associative. How many sets?

Q3. Calculation **Warning**
For a 64-bit byte address in the previous cache, find offset, set index and tag bits.

Q4. Calculation **Formula**
Hit time is 5 ns, main memory access is 100 ns and hit rate is 95 percent. Miss includes the hit lookup. Find AMAT.

Q5. MCQ
Which locality does sequential array traversal primarily exploit?

(A) Spatial locality
(B) Control hazard locality
(C) Opcode locality only
(D) Circuit switching

Q6. Short Answer **Strategy**
Explain compulsory, capacity and conflict misses.

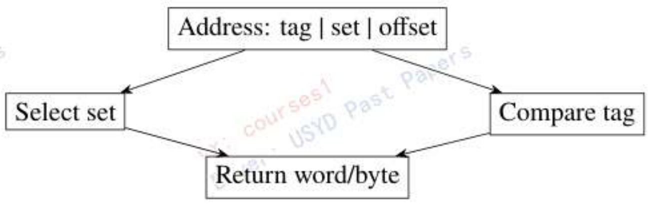
Q7. True/False
True or false: increasing associativity can reduce conflict misses but may increase hit-time complexity.

Q8. Short Answer **Pitfall**
Why can halving block size while keeping cache size constant change index and offset bits?

Q9. Table/Diagram
For a 32-bit byte address, a 1 KiB direct-mapped cache and 16-byte blocks, fill the address-field sizes:

tag	index	offset
?	?	?

Q10. Long Question
Consider this cache organisation.



(a) Explain the role of each field.
(b) Why is the tag comparison required?
(c) What changes in a 4-way set-associative cache?
(d) Give one cache-design trade-off.

W9 I/O, Storage and Amdahl's Law

Q1. Short Answer Pitfall

Why can I/O dominate total system performance even with a fast CPU?

Q2. MCQ

Which mechanism transfers large blocks between device and memory with limited CPU intervention?

(A) DMA
(B) De Morgan transform
(C) Immediate addressing
(D) Nonpersistent HTTP

Q3. Short Answer Strategy

Compare polling and interrupt-driven I/O.

Q4. Calculation Semester 1 2022 Final (modified)

Processes spend 60 percent on CPU and 40 percent on disk. CPU becomes 1.6 times faster. Find total speedup.

Q5. Calculation

With the same workload, disk becomes 1.9 times faster. Find total speedup.

Q6. Short Answer Strategy

Why should upgrade value sometimes be measured as improvement per dollar rather than raw speedup?

Q7. True/False

True or false: RAID 0 improves performance but provides no redundancy.

Q8. Short Answer Warning

Explain the write penalty idea in parity RAID.

Q9. Calculation

A component taking 45 percent of execution time is made 1.5 times faster. Compute overall speedup.

Q10. Long Question Semester 1 2024 Final (modified)

A server spends 70 percent of time in I/O and 30 percent in CPU. Option A improves I/O by 80 percent and costs 15000 dollars. Option B improves CPU by 50 percent and costs 10000 dollars.

(a) Compute speedup for A.
(b) Compute speedup for B.
(c) Which gives greater raw speedup?
(d) Which gives better improvement per dollar?

W10 Network Architecture and Layers

Q1. MCQ
Which layer is the best match for TCP in the five-layer Internet stack?

(A) Application
(B) Transport
(C) Network
(D) Data link

Q2. Short Answer Strategy
Match HTTP, SMTP, FTP, IP and Ethernet to their most appropriate layers.

Q3. Short Answer Novel
Explain encapsulation from application message to physical bits.

Q4. Calculation
A 50 Mbps circuit-switched link supports users needing 10 Mbps each. What is the maximum number of users?

Q5. Calculation Semester 1 2023 Final (modified)
In packet switching, 10 users are independently active with probability 0.3. What is the probability exactly one specified user is active and the other 9 are inactive?

Q6. True/False Pitfall
True or false: routers primarily forward packets based on network-layer information.

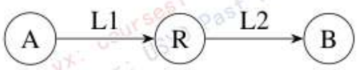
Q7. Short Answer
Why is packet switching efficient for bursty users?

Q8. MCQ
Which pair is most naturally associated with the application layer?

(A) HTTP and FTP
(B) IP and Ethernet
(C) TCP and UDP
(D) Fibre and copper

Q9. Short Answer Strategy
Give one advantage and one disadvantage of circuit switching.

Q10. Long Question
Two hosts communicate through a router:



(a) Name the network device in the middle.
(b) State which layers are mainly involved in end-host HTTP over TCP/IP.
(c) Explain what each link contributes to packet delay.
(d) Explain why the router must receive a packet before forwarding under store-and-forward.

W11 Delay, HTTP and Transport Headers

Q1. **Calculation** **Semester 1 2021 Final (modified)**
A 12000-bit packet crosses a 5 Mbps link of length 15 km with propagation speed 2×10^8 m/s. Compute tran propagation delay.

Q2. **Calculation** **Warning**
With two store-and-forward links, $L = 1000$ bits, $R_1 = 1$ Mbps, $R_2 = 500$ kbps, and total propagation delay 3 ms, find total delay ignoring queue/processing.

Q3. **Short Answer** **Formula**
Explain why propagation delay and transmission delay are different.

Q4. **Calculation**
A web page has one HTML file and six small referenced objects. Nonpersistent HTTP with no parallelism needs how many RTTs?

Q5. **Calculation**
For the same page, nonpersistent HTTP with at most three parallel TCP connections for referenced objects needs how many RTTs?

Q6. **Calculation**
For the same page, persistent HTTP without pipelining including TCP setup needs how many RTTs?

Q7. **MCQ** **Pitfall**
Which field is not part of the basic UDP header?

(A) Source port
(B) Destination port
(C) Length
(D) Cache tag

Q8. **Calculation**
A UDP header length field is 0x0020. What is the total datagram length in bytes and the data length in bytes?

Q9. **Short Answer** **Formula**
How is an Internet checksum computed at a high level?

Q10. **Long Question**
A browser requests a small HTML page referencing m small same-server objects. Ignore DNS and transmission time.

(a) Derive RTT count for nonpersistent HTTP without parallelism.
(b) Derive RTT count for nonpersistent HTTP with c parallel connections after the HTML file.
(c) Derive RTT count for persistent HTTP without pipelining.
(d) Apply the formulas to $m = 6, c = 3$.

W12 Web Cache and Integrated Performance

Q1. Short Answer Strategy
What roles does a web cache play when serving a browser request?

Q2. Calculation
Average object size is 100 Kbits and request rate is 15/s over a 1.54 Mbps access link. Compute utilization.

Q3. MCQ Warning
Why is access-link utilization near 1 dangerous?

(A) Queueing delay can grow dramatically.
(B) It changes binary to decimal.
(C) It removes all propagation delay.
(D) It makes TCP an application protocol.

Q4. Calculation
A cache hit delay is 20 ms, origin delay is 2 s and hit rate is 0.6. Estimate average delay ignoring access-link interactions.

Q5. Short Answer Strategy
Give two ways a web cache improves perceived performance.

Q6. True/False
True or false: a proxy cache can be both client and server depending on direction of communication.

Q7. Short Answer
Why might simply increasing access bandwidth be less attractive than caching?

Q8. Calculation Pitfall
A component with fraction $f = 0.8$ is made infinitely fast. What is the maximum total speedup?

Q9. Short Answer
List five final-exam checks before submitting a COMP9601 calculation.

Q10. Long Question
An institution considers either installing a web cache or upgrading its access link.

(a) State the main performance bottleneck when utilization is 97 percent.
(b) Explain how hit rate affects origin traffic.
(c) Give one case where bandwidth upgrade still matters.
(d) Relate this decision to the general bottleneck principle from computer organisation.

Answer Key

W1 Intro and System View

1. **Q1.** B. The stored-program model stores both instructions and data as addressable bit patterns.
2. **Q2.** Latency is time for one operation; throughput is operations per unit time. A faster CPU may increase throughput only if jobs are CPU-bound; if the disk dominates, single-job latency and throughput may barely improve.
3. **Q3.** $2.5 \times 10^6/(0.5 \times 10^{-3}) = 5 \times 10^9$ instr/s = 5000 MIPS.
4. **Q4.** False. It can address 2^{32} bytes. Addressable words depend on word size/alignment.
5. **Q5.** Because exam questions mix bits/bytes, Mbps/bps, ms/ns, words/bytes. Unit mistakes can change answers by factors of 8 or powers of 10/2.
6. **Q6.** B. Waiting for storage indicates an I/O or memory bottleneck.
7. **Q7.** That details inside each block can be abstracted at the current level, and that the main interactions are instruction execution, storage/retrieval and external data transfer.
8. **Q8.** 8 bytes and 16 nibbles.
9. **Q9.** If it lacks the formula, units, assumptions or interpretation requested by the question.
10. **Q10.** Memory and I/O are missing. If 80 percent of time is disk waiting, even a very fast CPU changes only the CPU fraction. Latency can be seconds/request; throughput can be requests/s or instructions/s. Amdahl formalises that accelerating only a fraction limits total speedup.

W2 Radix and Integer Representation

1. **Q1.** $187 = 2 \cdot 81 + 0 \cdot 27 + 2 \cdot 9 + 2 \cdot 3 + 1$, so $(20221)_3$.
2. **Q2.** $(34)_8 = 3 \cdot 8 + 4 = 28_{10} = 1 \cdot 16 + 3 \cdot 4 + 0 = (130)_4$.
3. **Q3.** C. $25 = 00011001$; invert/add 1 gives 11100111.
4. **Q4.** Because the sign bit can be positive or negative while the magnitude/all remaining bits are zero.
5. **Q5.** Minimum $-2^{11} = -2048$, maximum $2^{11} - 1 = 2047$.
6. **Q6.** False. Unsigned carry and signed overflow are different tests; signed overflow depends on operand/result signs.
7. **Q7.** Truncation loses low-significance information when representation is shortened; overflow occurs when the value is outside the representable range.
8. **Q8.** $2^{20} \times 2 = 2^{21}$ bytes.
9. **Q9.** 0x4B at 0x134A, 0xA7 at 0x134B, 0x03 at 0x134C, 0x00 at 0x134D.
10. **Q10.** $+17 = 010001$. $25 = 011001$, so $-25 = 100111$. Sum $010001 + 100111 = 111000$, which is -8. No overflow because operands have different signs.

W3 Floating Point and Error Codes

1. **Q1.** Only finitely many fractions are representable, so real values are rounded to nearby representable values; repeated operations can accumulate error.
2. **Q2.** $2^{5-1} - 1 = 15$.
3. **Q3.** B. CRC is designed for strong error detection, including many burst-error patterns.
4. **Q4.** The data has four 1s, already even, so append 0.
5. **Q5.** CRC arithmetic is over GF(2), so addition and subtraction are both XOR with no carries or borrows.
6. **Q6.** Three zeros, because generator 1001 has degree 3.
7. **Q7.** True.
8. **Q8.** Parity can detect some errors but usually cannot locate/correct them. Hamming checks create a syndrome that can identify and correct a single-bit error.
9. **Q9.** 10 explicit fraction bits; the leading 1 is not stored for normal numbers.
10. **Q10.** Degree 3; append 000 before modulo-2 division; append the computed 3-bit remainder to the original data; receiver divides the full received bit string by 1001. Nonzero remainder means a detected error, not necessarily its location.

W4 Boolean Algebra and Logic Gates

1. **Q1.** 0,0,0,0,0,1,1,1.
2. **Q2.** It simplifies to x by absorption.
3. **Q3.** C. De Morgan's theorem.
4. **Q4.** Either can be combined to implement NOT, AND and OR; therefore they can implement any Boolean function.

5. **Q5.** $y + y' = 1$, so $x(1) + xz = x + xz = x$.
6. **Q6.** True. OR gives 1 for 11; XOR gives 0.
7. **Q7.** Eight rows because there are three Boolean variables.
8. **Q8.** Label every intermediate wire, write each gate's output expression locally, then substitute from inputs
9. **Q9.** SOP ORs product terms/minterms, naturally built from rows where the function is 1. POS ANDs sum terms/maxterms, naturally built from rows where the function is 0.
10. **Q10.** $F = x(yz + y + z)$. Since $y + yz = y$, $F = x(y + z)$. For example $x, y, z = 1, 1, 0$ gives 1. A truth table compares the original and simplified outputs for all eight possible inputs.

W5 Simple Computer and Stack Expressions

1. **Q1.** B. $\lceil \log_2 75 \rceil = 7$.
2. **Q2.** Opcode bits $= \lceil \log_2 61 \rceil = 6$; address bits $= 10$.
3. **Q3.** PC stores next instruction address; IR stores current instruction; MAR stores memory address for access; MBR/MDR stores data moving to/from memory.
4. **Q4.** AB+CDE/-*.
5. **Q5.** $(A + B) * (C - (D/E))$.
6. **Q6.** True.
7. **Q7.** Because A^{B^C} groups as $A^{(B^C)}$, so the pop rule differs from left-associative operators.
8. **Q8.** $2^{12} = 4096$ words.
9. **Q9.** Opcode selects the operation; operand/address fields identify data, registers, constants or targets used by that operation.
10. **Q10.** $2^4 = 16$. LOAD 0x400 is binary 0001 0100 0000 0000=0x1400. HALT is 1111 0000 0000 0000=0xF000. Hex groups exactly into 4-bit nibbles, matching the opcode and address nibbles.

W6 Instruction Processing and Machine Code

1. **Q1.** Fetch instruction from memory using PC, load IR, increment/update PC, decode opcode/addressing mode, fetch operands if needed, execute, write result.
2. **Q2.** B. I/O readiness commonly triggers an interrupt.
3. **Q3.** 0x3402.
4. **Q4.** Opcode bits are 0101 and the address field is 0x100. If the course opcode table defines 0101 as JMP, then the instruction is JMP 0x100.
5. **Q5.** SUB LIMIT overwrites AC with X-LIMIT and sets the condition used for the branch. The program should branch/test immediately before another instruction overwrites AC or condition state; reload X later if needed.
6. **Q6.** 55.
7. **Q7.** True.
8. **Q8.** Labels avoid off-by-one address mistakes while the instruction sequence is still changing; addresses can be resolved after layout.
9. **Q9.** 0x109 if the first instruction is 0x100 and there are 10 consecutive instructions.
10. **Q10.** Use SUM=0, X=1, ONE=1, LIMIT=11. Repeatedly load SUM, add X, store SUM; load X, add ONE, store X; subtract LIMIT; if zero exit else jump to loop. LIMIT=11 exits after X has been incremented beyond 10. Final SUM=55.

W7 ISA, Addressing Modes and Pipelining

1. **Q1.** B. RISC designs tend to use simple regular instructions that suit pipelining.
2. **Q2.** Effective address = 500; loaded value = Memory[500] = 200.
3. **Q3.** Indirect uses Memory[Memory[500]]=Memory[200]=900.
4. **Q4.** Effective address is 500+300=800; value depends on Memory[800].
5. **Q5.** The operand field is the literal value. It avoids an extra memory access for constants.
6. **Q6.** False. It primarily improves throughput after the pipeline fills.
7. **Q7.** $(4 + 5 - 1) \times 2 = 16$ ns.
8. **Q8.** Structural, data and control hazards. Example: data forwarding can reduce stalls for data hazards.
9. **Q9.** Variable length and more complex memory/operand behaviour make stage timing and decode less regular.
10. **Q10.** Without pipelining $20 \times 4 \times 3 = 240$ ns. Pipelined time $(4 + 20 - 1) \times 3 = 69$ ns. Speedup ≈ 3.48. Hazards, imbalance and branch stalls reduce real speedup.

W8 Memory Hierarchy and Cache

1. **Q1.** Offset $= \log_2 64 = 6$; line $= \log_2 256 = 8$; tag $= 30 - 8 - 6 = 16$.
2. **Q2.** $4\text{MiB}/(512 \times 16) = 2^{22}/2^{13} = 2^9 = 512$ sets.
3. **Q3.** Offset 9, set index 9, tag $64 - 9 - 9 = 46$.
4. **Q4.** $0.95(5) + 0.05(105) = 10$ ns, equivalently $5 + 0.05(100) = 10$ ns.
5. **Q5.** A. Nearby addresses are used soon.
6. **Q6.** Compulsory: first reference; capacity: working set too large for cache; conflict: mapping/associativity forces eviction despite enough total capacity.
7. **Q7.** True.
8. **Q8.** Offset bits decrease because each block is smaller; number of blocks/sets increases, so index bits may increase depending on associativity.
9. **Q9.** Offset $= 4$; cache lines $= 1024/16 = 64$, so index $= 6$; tag $= 32 - 6 - 4 = 22$.
10. **Q10.** Offset selects byte/word within block; set/index chooses candidate set; tag identifies which memory block is actually present. Tag comparison prevents returning data from a different block that maps to the same set. Four-way cache compares up to four tags in the selected set and uses replacement on miss. Trade-off: larger/associative caches reduce misses but may increase hit time, cost and energy.

W9 I/O, Storage and Amdahl's Law

1. **Q1.** I/O devices are much slower and may force the CPU or process to wait; the large waiting fraction limits total speedup.
2. **Q2.** A. Direct memory access.
3. **Q3.** Polling repeatedly checks device status and wastes CPU if waits are long. Interrupts let the device signal when service is needed.
4. **Q4.** $S = 1/(0.4 + 0.6/1.6) = 1/0.775 = 1.29$.
5. **Q5.** $S = 1/(0.6 + 0.4/1.9) = 1.234$.
6. **Q6.** Budgets matter; a smaller speedup may be better value if it costs much less.
7. **Q7.** True.
8. **Q8.** Updating data may require reading old data/parity and writing new data/parity, adding extra I/O compared with simple writes.
9. **Q9.** $1/(0.55 + 0.45/1.5) = 1/0.85 = 1.176$.
10. **Q10.** A: $1/(0.3 + 0.7/1.8) = 1.452$. B: $1/(0.7 + 0.3/1.5) = 1.111$. A gives greater raw speedup. Improvement per dollar: A $0.452/15000$, B $0.111/10000$; A is also better.

W10 Network Architecture and Layers

1. **Q1.** B. TCP is a transport-layer protocol.
2. **Q2.** HTTP/SMTP/FTP are application; IP is network; Ethernet is data link in the usual five-layer exam model.
3. **Q3.** Each lower layer wraps the higher-layer payload with its own header/trailer: message, segment, datagram, frame, bits.
4. **Q4.** 5 users.
5. **Q5.** $0.3(0.7)^9$.
6. **Q6.** True.
7. **Q7.** It statistically multiplexes users so idle users do not reserve bandwidth.
8. **Q8.** A. HTTP and FTP are application protocols.
9. **Q9.** Advantage: reserved predictable bandwidth. Disadvantage: inefficient when users are idle or bursty.
10. **Q10.** The middle device is a router. HTTP is application, TCP transport, IP network, link/physical on each hop. Each link contributes transmission delay and propagation delay. Store-and-forward requires the full packet to arrive so the router can check/process and transmit it on the outgoing link.

W11 Delay, HTTP and Transport Headers

1. **Q1.** Transmission $= 12000/(5 \times 10^6) = 2.4$ ms. Propagation $= 15000/(2 \times 10^8) = 0.075$ ms.
2. **Q2.** 1 ms $+2$ ms $+3$ ms $= 6$ ms.
3. **Q3.** Propagation is time for a signal to travel distance; transmission is time to push all packet bits onto a link.
4. **Q4.** Seven objects times 2 RTT each = 14 RTT.
5. **Q5.** HTML first costs 2 RTT. Six objects are fetched in two waves, each 2 RTT, so total 6 RTT.

6. **Q6.** One TCP setup RTT plus one request/response RTT for each of 7 objects = 8 RTT.
7. **Q7.** D. Cache tag is not a UDP header field.
8. **Q8.** 0x0020 = 32 bytes total. UDP header is 8 bytes, so data length is 24 bytes.
9. **Q9.** Split data into 16-bit words, add with one's-complement wraparound carry, then complement the final sum.
10. **Q10.** Nonparallel nonpersistent: $2(1+m)$. Parallel: $2+2\lceil m/c \rceil$. Persistent no pipelining: $1+(1+m)$. For $m=6, c=3$: 14 RTT, 6 RTT and 8 RTT.

W12 Web Cache and Integrated Performance

1. **Q1.** It acts as a server to the browser. On a miss it also acts as a client to the origin server and may store the returned object.
2. **Q2.** $\rho = \lambda L/R = 15 \times 100000/1540000 = 0.974$, about 97.4 percent.
3. **Q3.** A. Queueing delay becomes very large as utilization approaches 1.
4. **Q4.** $0.6(0.02) + 0.4(2) = 0.812$ s.
5. **Q5.** It serves hits locally, reducing response time, and reduces traffic over the access link, lowering congestion/queueing.
6. **Q6.** True.
7. **Q7.** Bandwidth upgrades can be costly and still leave repeated origin RTTs; caching directly removes many origin fetches and reduces load.
8. **Q8.** $1/(1-f) = 5$.
9. **Q9.** Check units, powers of two/ten, bit fields sum to address width, object count/RTT assumptions, and whether the answer needs interpretation.
10. **Q10.** The access link queue is the bottleneck. Hit rate reduces origin traffic roughly by the miss fraction, lowering access utilization and delay. Bandwidth still matters for misses, large uncached objects and low locality. This is the same bottleneck principle as Amdahl: improving the dominant fraction gives the largest total effect.